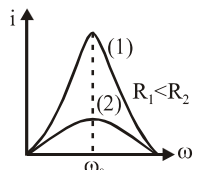


HINTS & SOLUTION

1. Chemical and magnetic effect depends on direction of current but heating effect not depends on direction of flow of current.
2. A signal appears across that element for which it comes across high impedance.
3. The average value of a.c. is zero because it oscillates symmetrically between positive and negative values.
4. $P_{av} = V_{rms} I_{rms} \cos \phi$, when power factor ($\cos \phi$) is low then I_{rms} is large. Since power transmitted at fixed voltage, with large I_{rms} , ($P_{loss} = I_{rms}^2 R$) is high.
5. $Q = \frac{X_L}{R} = \frac{L\omega}{R}$
6. When we walk through metal detector with metal the impedance changes then current also changes in circuit, then cause a sound.
10. i_{rms} = effective value of current
11. Phase is R way of represent, phases of current and voltage is AC.
12. $\frac{di}{dt} = \frac{V_m}{L} \sin \omega t$
 $i = \frac{V_m}{\omega L} (-\cos \omega t)$
13. For resonance circuit must have internal frequency.
14. Resonance and quality factor
15. At Resonance amplitude becomes maximum.
16. When external frequency = circuit frequency, then circuit is at resonance and $X_L = X_C$ and
 $i = \frac{V}{R}$.
17. At half power frequency current becomes $\frac{1}{\sqrt{2}}$.
18. $\omega = \frac{1}{\sqrt{LC}}$
20.  $Q = \frac{X_L}{R} = \frac{\omega L}{R}$
21. Damping always occurs practically.
22. $\omega \uparrow X_L \uparrow X_C \downarrow$